

This assignment will **NOT** be graded and your solutions need not be submitted. It is intended to help you review some of the probability background you may need for this course. The below problems are listed roughly in order of increasing difficulty.

1. Do problems 1.3, 1.6, 1.10, 2.8, 2.27 from Bishop, Pattern Recognition and Machine Learning (copied below).

- **1.3** Suppose that we have three coloured boxes r (red), b (blue), and g (green). Box r contains 3 apples, 4 oranges, and 3 limes. Box b contains 1 apple, 1 orange, and 0 limes. Box g contains 3 apples, 3 oranges, and 4 limes. If a box is chosen at random with probabilities $p(r) = 0.2$, $p(b) = 0.2$, and $p(g) = 0.6$ and a piece of fruit is removed from that box with uniform probability, then what is the probability of selecting an apple? If we observe that the selected fruit is an orange, what is the probability that it came from the green box?
- **1.6** Show that if two variables x and y are independent, then their covariance $Cov(x, y) = 0$.
- **1.10** Suppose that the two variables x and z are statistically independent. Show that the mean μ and variance σ^2 of their sum $(x + z)$ satisfies:

$$\begin{aligned}\mathbb{E}[x + z] &= \mathbb{E}[x] + \mathbb{E}[z] \\ \text{Var}[x + z] &= \text{Var}[x] + \text{Var}[z]\end{aligned}$$

- **2.8** Consider the two variables x and y with joint distribution $p(x, y)$. Prove the following results:

$$\begin{aligned}\mathbb{E}[x] &= \mathbb{E}_y[\mathbb{E}_x[x|y]] \\ \text{Var}[x] &= \mathbb{E}_y[\text{Var}_x[x|y]] + \text{Var}_y[\mathbb{E}_x[x|y]]\end{aligned}$$

Here $\mathbb{E}_x[x|y]$ denotes the expectation of x under the conditional distribution $p(x|y)$, with a similar notation for the conditional variance.

- **2.27** Let $\mathbf{x}$ and $\mathbf{z}$ be two independent random vectors, so that $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{x})p(\mathbf{z})$.
 - Show that the mean of their sum $\mathbf{y} = \mathbf{x} + \mathbf{z}$ is given by the sum of the means of each of the variables separately.
 - Show that the covariance matrix of $\mathbf{y}$ is given by the sum of the covariance matrices of $\mathbf{x}$ and $\mathbf{z}$.

Confirm that these results agree with **1.10** (above).

2. Suppose the real, scalar random variables x , y , and z are jointly Gaussian. Are each of the following Gaussian, not Gaussian, or unknown (i.e. could be either Gaussian or non-Gaussian depending on specifics of the distribution)?

- x
- $2x + y - z$
- x^2
- $x|y$
- $x|y, z$

3. Two scalar random variables x and y are known to be uncorrelated. Show that x and y need not be independent, by defining (implicitly or explicitly) a joint distribution for x and y such that they are uncorrelated but not independent. Does your answer change if x and y are jointly Gaussian?